AI ACADEMY · LEVEL 2 · GRADE 7 · WEEK 10 · STUDENT LESSON

Human Classifier Simulation

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

Check the input. Follow the branch. Keep the outcome.

Build a partner-operable paper classifier with explicit rejection, versioned records and an honest evidence limit.

Keep one habit: Supplied Tower1 rule → partner trace → separate eligibility filter → independent strict boundary.

MAIN LESSON · SECTION 1

Can a partner follow your classifier without guessing?

> Write input checks, an explicit rejection path and the category rule.

Your learning goals

I can construct a **classifier** with a stated **input** format, selected rule and recorded **output**.

I can trace valid, **boundary** and invalid **input**s through different paths.

I can distinguish an **input** check from **evidence** that a **prediction** is correct.

Check equality and missing input

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

- A supplied rule predicts tall at a height of at least 6 cm and short otherwise. Predict at 5 and 6 cm. What should you check before applying it to a blank height?

The finished procedure you will produce

By the end, I can simulate a **classifier** with explicit requirements and missing-value behavior.

▼ **Earlier wording, alternatives and picture transcripts**

Your learning target

Remember before reading

Without opening the old lesson, explain one key idea from Features and Distractors. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Rows Columns and Data Types. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Trace Tables. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 10: Human **Classifier** Simulation

A **classifier** applies a learned or designed **boundary** to features.



THINKER MISSION

Train classmates on labeled creatures and compare their rules.

Thinker name and date:

Write or draw your reasoning and evidence.

Readiness for the connected explanation

This week’s end goal

MAIN LESSON · SECTION 2

Start with a supplied tower rule and an input contract

> A threshold alone is not a complete operating procedure.

The supplied Tower1 rule

Your task is to build a paper **classifier** that another learner can operate without guessing your instructions. It will predict short or tall for imaginary towers. For this week, use the supplied rule: predict tall at a height of at least 6 centimetres and short otherwise. Name this rule Tower1. We will investigate selecting rules from examples later; no fitting experiment has been performed here.

What inputs this version accepts

Write an **input contract**: one height recorded as a whole number of centimetres from 1 through 10, including both endpoints. Require the unit cm to be shown. This is a classroom design choice about what the procedure accepts.

It is not a claim that every accepted height appeared in **fitting** or that towers outside this range cannot exist.

Why both acceptance and rejection paths matter

A complete **classifier** needs instructions for checking the **input**, applying Tower1 when the **input** is accepted, and recording the outcome. It must also say what happens when the **input** fails a check.

The threshold alone does not provide these surrounding instructions.

▼ Earlier wording, alternatives and picture transcripts

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



MYSTERY

The learners must use human **classifier** simulation to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

> **Write or draw your reasoning and evidence.**

My **prediction** and the clue supporting it:

> **Write or draw your reasoning and evidence.**

A closer explanation

MAIN LESSON · SECTION 3

Check first; compare only accepted heights

> Stop processing a rejected record before applying the threshold.

The check-and-stop instruction

For every record, first check that a height is present and its stated unit is cm. Then check that it is a whole number from 1 through 10.

If any check fails, record **no prediction** and the reason, then stop **process**ing that record. Do not continue into the threshold comparison.

Trace 5 cm, 6 cm and a blank

AI ACADEMY / AI-2/10 · 03 · INSPECT THE LESSON MODEL

Three records through the Tower1 classifier

Input record	Check and rule	Recorded outcome
5 cm	Accepted; 5 is below 6	Tower1 predicts short
6 cm	Accepted; 6 is at least 6	Tower1 predicts tall
Height blank	Missing height; stop before rule	Tower1; no prediction

Read each row from left to right. An accepted input reaches the threshold rule. A missing input stops before that rule and receives an explicit no prediction outcome.

Follow the labelled picture with the matching explanation. Open the vector diagram

Record original input, version and outcome

If all checks pass, compare the height with 6. At least 6 gives the **prediction** tall; otherwise give short. Record the original **input**, **model version** Tower1 and the **prediction**.

For a rejected record, record the original **input**, Tower1, **no prediction** and a reason. Tower1 identifies the intended model even when its threshold was not used.

Explain the branch in the diagram

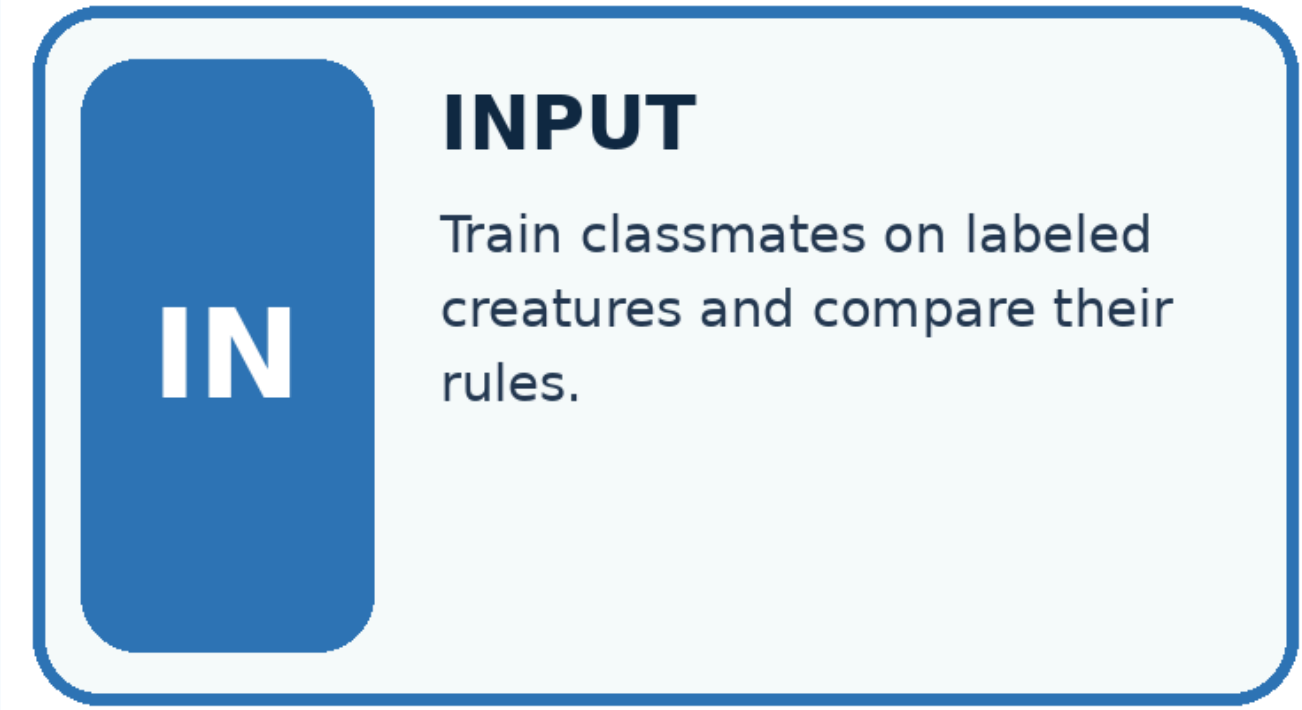
- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ **Earlier wording, alternatives and picture transcripts**

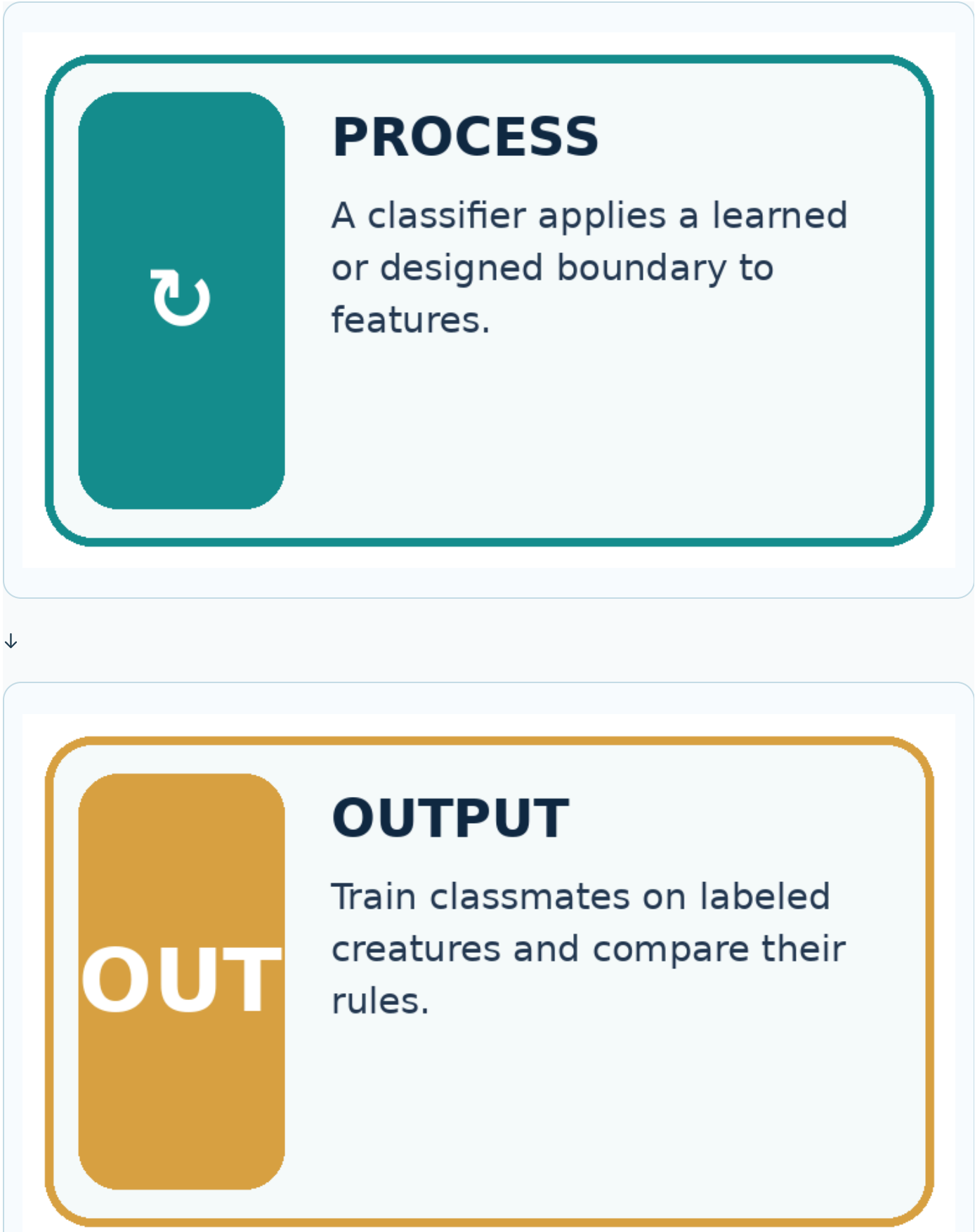
VISUAL **MECHANISM**

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.



↓


↻
PROCESS
A classifier applies a learned or designed boundary to features.
↓
OUT
OUTPUT
Train classmates on labeled creatures and compare their rules.

Which stage could fail even when the other two are correct? Explain:

> **Write or draw your reasoning and evidence.**

A closer explanation

Three records through the Tower1 classifier

INPUT RECORD	CHECK AND RULE	RECORDED OUTCOME
5 cm	Accepted; 5 is below 6	Tower1 predicts short
6 cm	Accepted; 6 is at least 6	Tower1 predicts tall
Height blank	Missing height; stop before rule	Tower1; no prediction

Read each row from left to right. An accepted input reaches the threshold rule. A missing input stops before that rule and receives an explicit no prediction outcome.

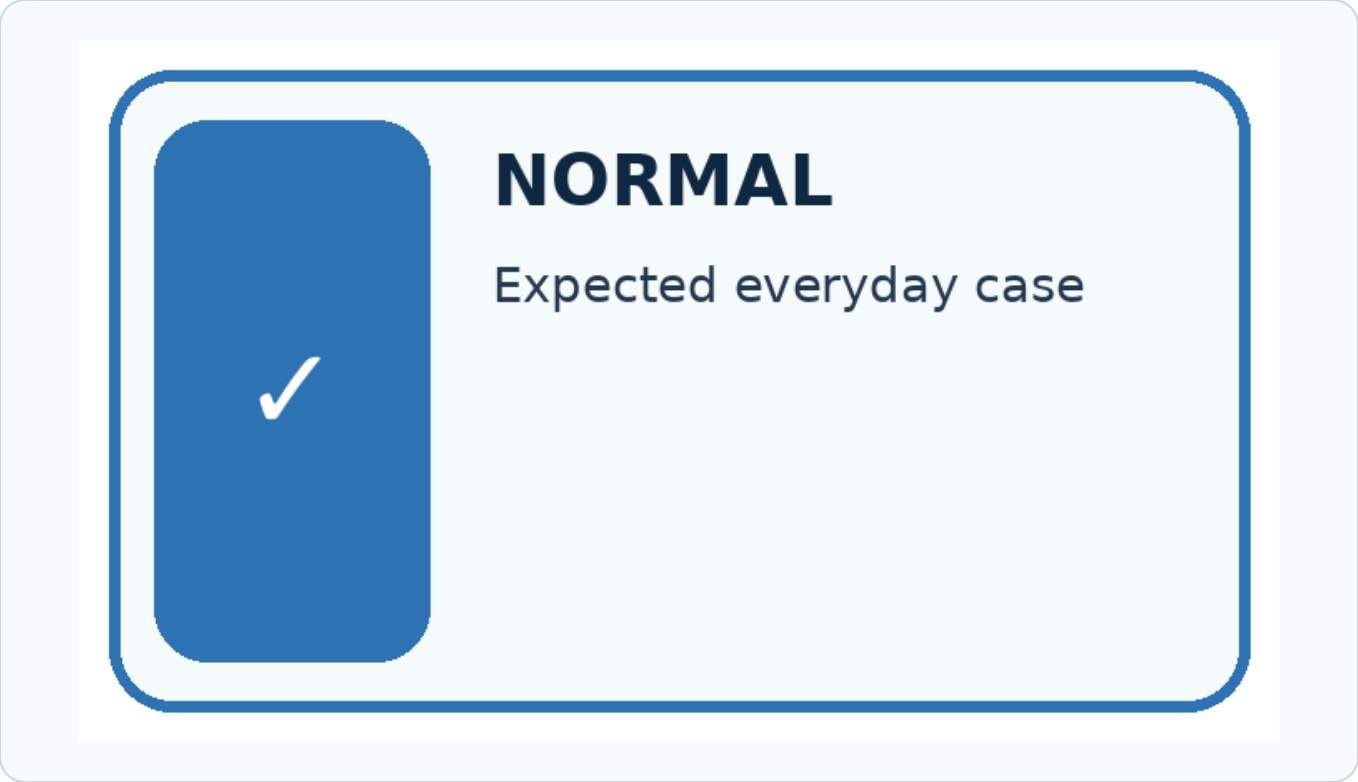
Connect the picture and the explanation

MAIN LESSON · SECTION 4

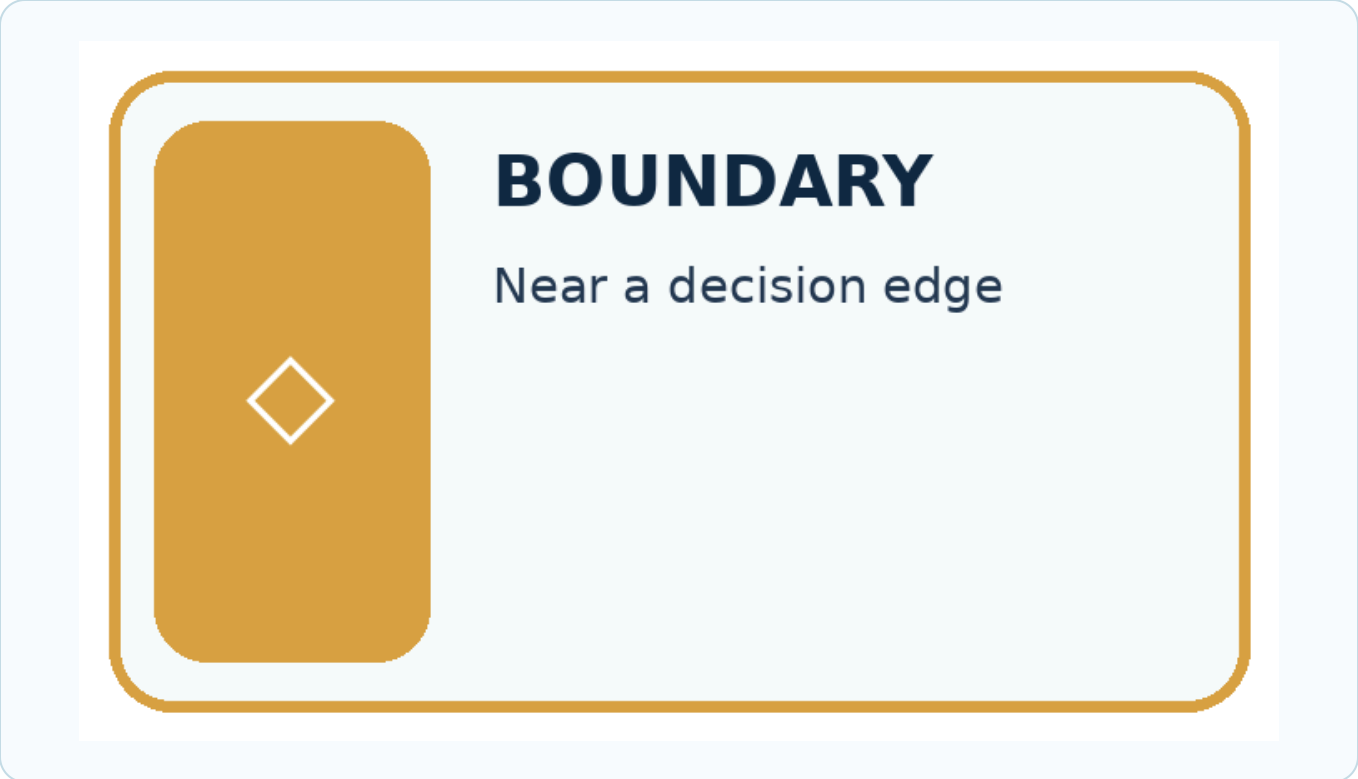
Challenge endpoints, equality and missing evidence

> Keep contract rejection distinct from a predicted category.

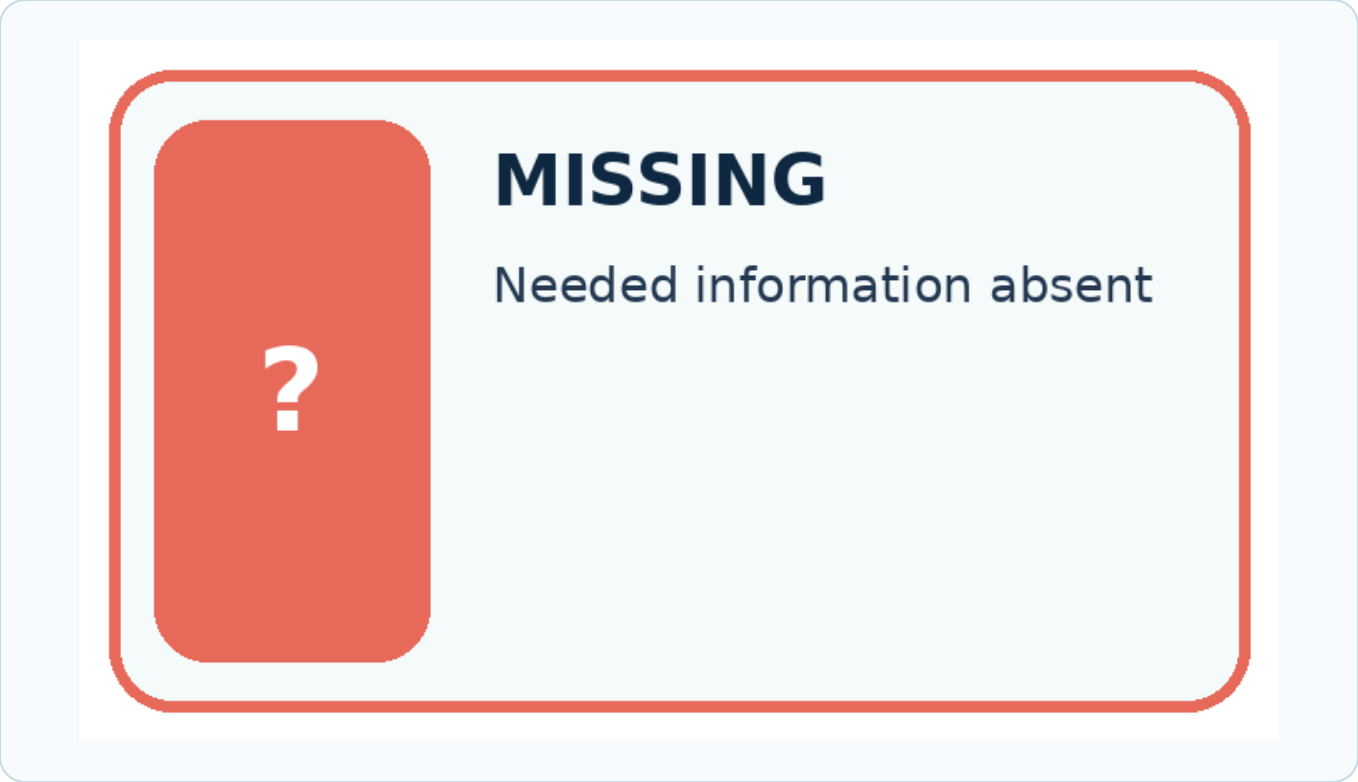
Four pictures with a purpose



Normal case: predict the result and deciding evidence.



Boundary case: identify the exact limit.



Missing information: name what must be clarified.



Transfer case: explain which procedure still applies.

Choose a revealing case

- Which case is most likely to reveal a hidden assumption? Why?

> **Write or draw your reasoning and evidence.**

Predict the supplied changed case

- Change: Supply M's pages as 101.

▼ Check after predicting

Prediction: M is now rejected rather than awaiting page review.

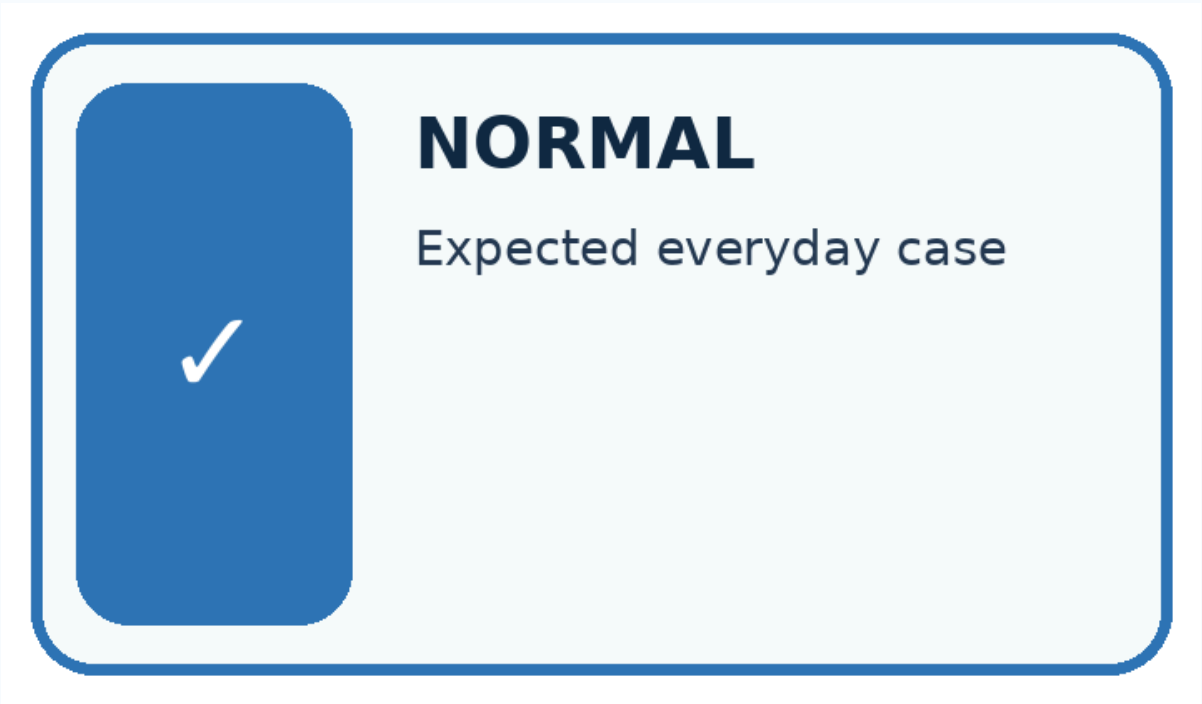
Reason: The required field is known but fails the maximum-length condition.

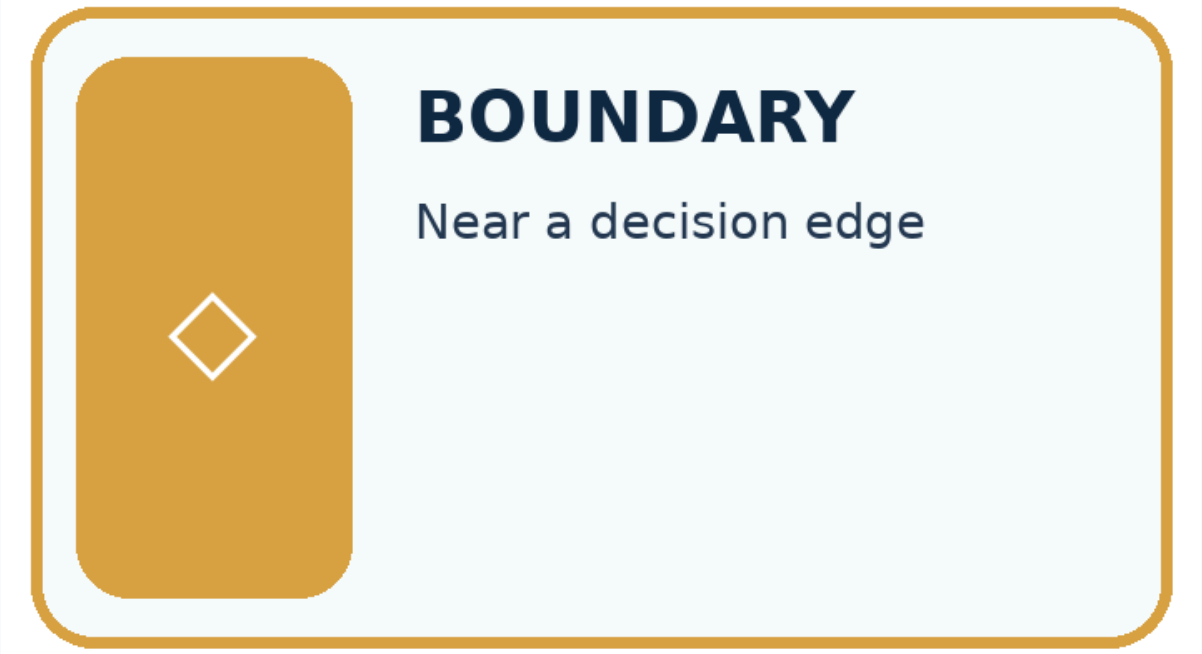
▼ Earlier wording, alternatives and picture transcripts

CASE GALLERY

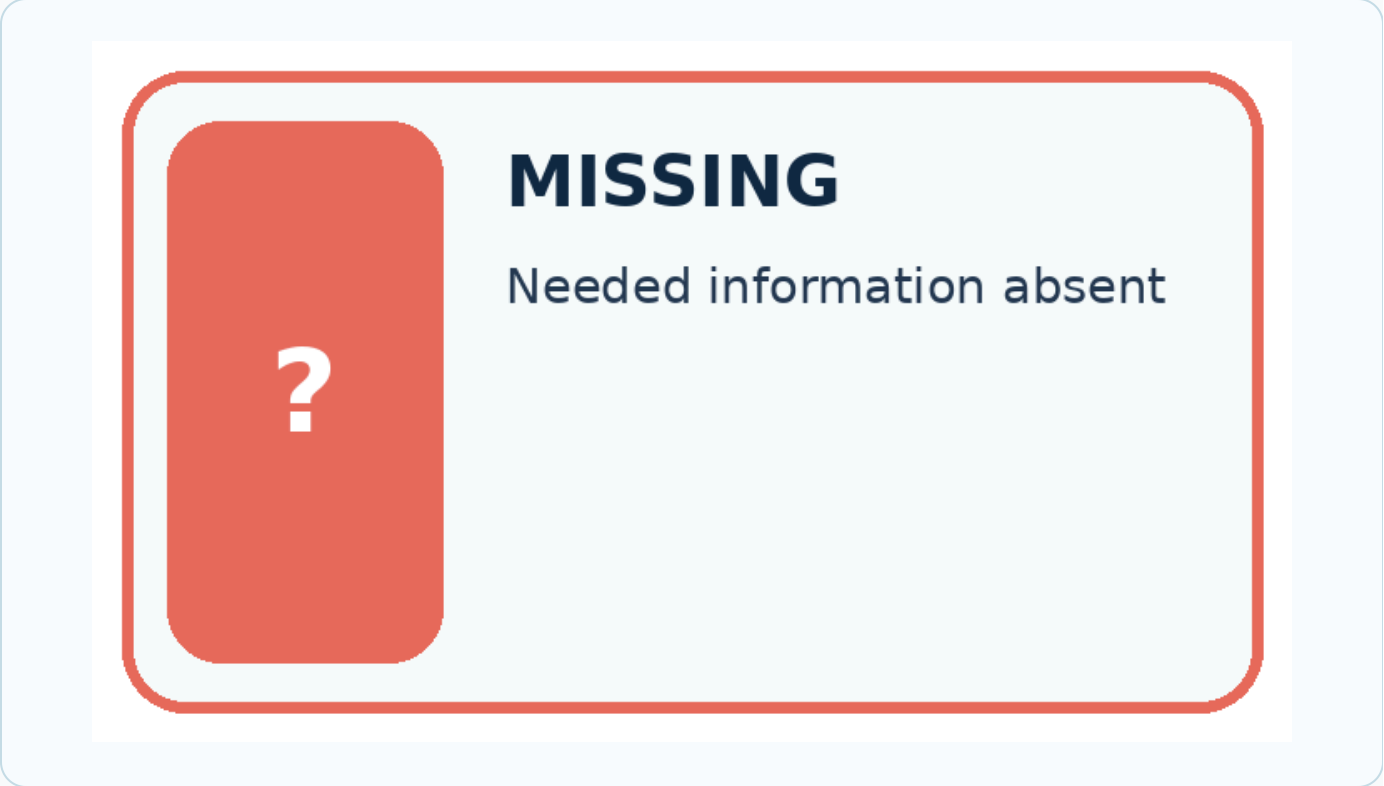
Four Tests, Four Kinds of **Evidence**

A system is not understood until it survives more than one comfortable example.





NORMAL CASE: predict and name **evidence**. **BOUNDARY** CASE: predict and name **evidence**.





MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Trace Tower1, then a separate book-eligibility rule

> The book example has three requirements rather than a height threshold.

Predict the accepted and rejected Tower1 cases

Input	Contract result	Outcome
1 cm	Accepted at lower endpoint	Predict short
9 cm	Accepted	Predict tall
10 cm	Accepted at upper endpoint	Predict tall
0 cm	Outside 1 through 10	No prediction
60 mm	Unit is not cm	No prediction
4.5 cm	Not a whole number	No prediction

Separate contract validation from correctness

A valid 9 cm **input** receives a tall **prediction**, but the **input** check does not establish that the **prediction** matches its comparison label. **Validation** checks the contract.

Evaluation checks **prediction**s against appropriate **evidence**. Keep these questions separate.

Explain why 60 mm is rejected in this version

Sixty millimetres equals six centimetres, but this version accepts cm records only. It does not contain a conversion step.

Ask for a correctly recorded cm measurement or deliberately design and check a conversion step in a new version. Do not silently treat the digits 60 as centimetres. A missing height must not become an invented zero either.

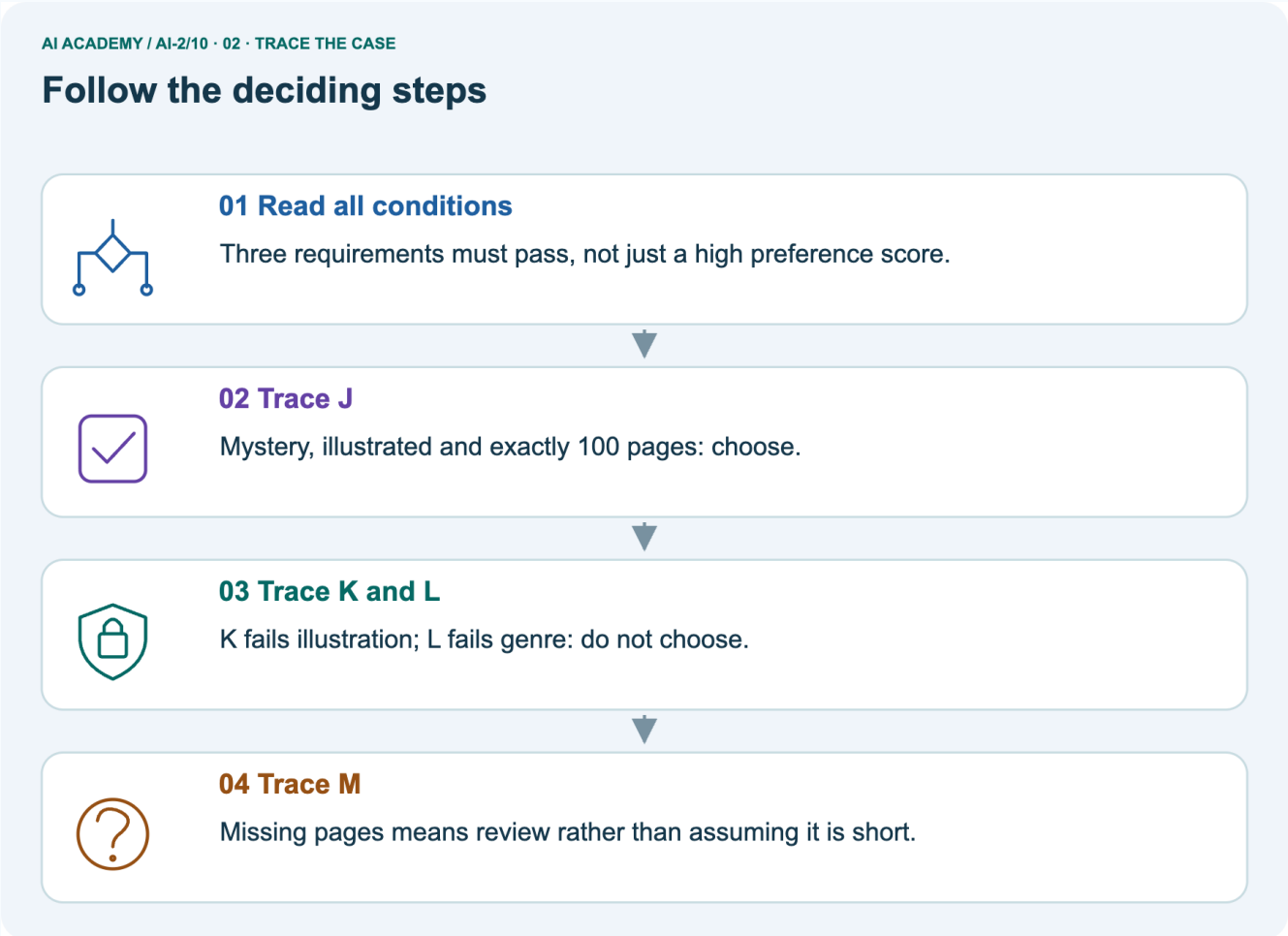
Introduce the illustrated-mystery contract

A fictional librarian wants illustrated mystery books with at most 100 pages.

Written rule: choose only genre mystery AND illustrated True AND pages<=100.

Rows J=(mystery,True,100), K=(mystery,False,90), L=(science,True,80), M=(mystery,True,blank). Missing required fields request review.

Trace J, K, L and missing-pages M



Follow the labelled picture with the matching explanation. Open the vector diagram

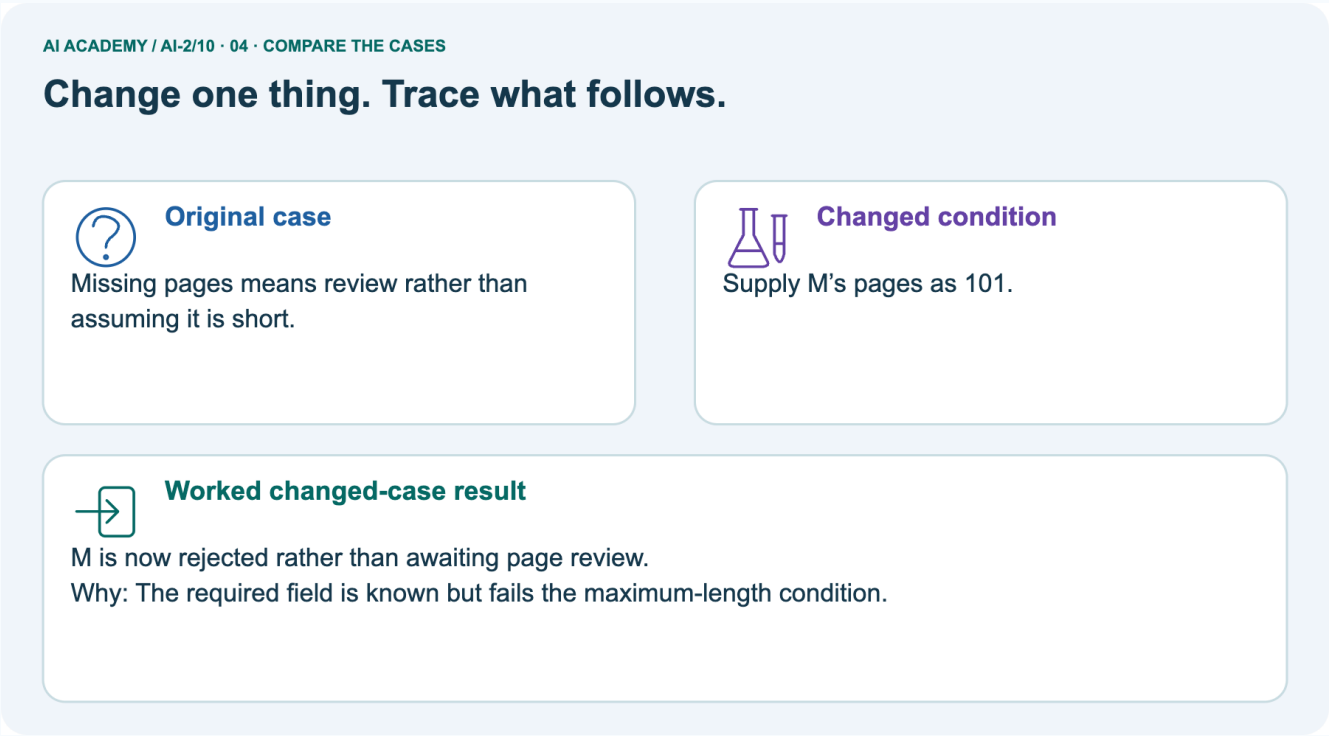
This human simulation lets students inspect every decision without needing a trained model. Equality matters at 100. Known failures give reasons for rejection; M lacks enough information for the page check.

The order should not turn blank into zero or quietly ignore a required field.

Predict M after the missing value is supplied

- Change one thing: Supply M's pages as 101. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain the missing field and the eligibility limit

- Read the labelled picture: Why are J and M initially treated differently even though both are illustrated mysteries?

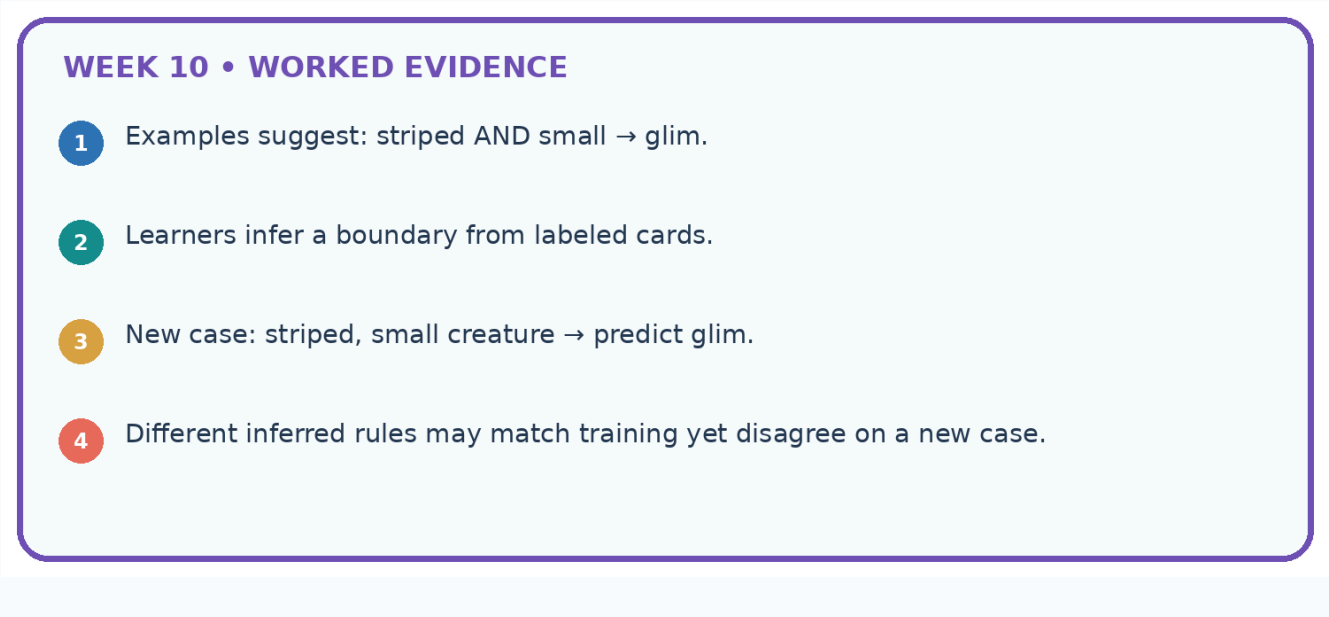
- This declared rule checks eligibility, not whether an eligible book will actually be enjoyed.

▼ Earlier wording, alternatives and picture transcripts

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.



1. Examples suggest: striped AND small → glim.

2. Learners infer a **boundary** from labeled cards.

3. New case: striped, small creature → predict glim.

4. Different inferred rules may match **training** yet disagree on a new case.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

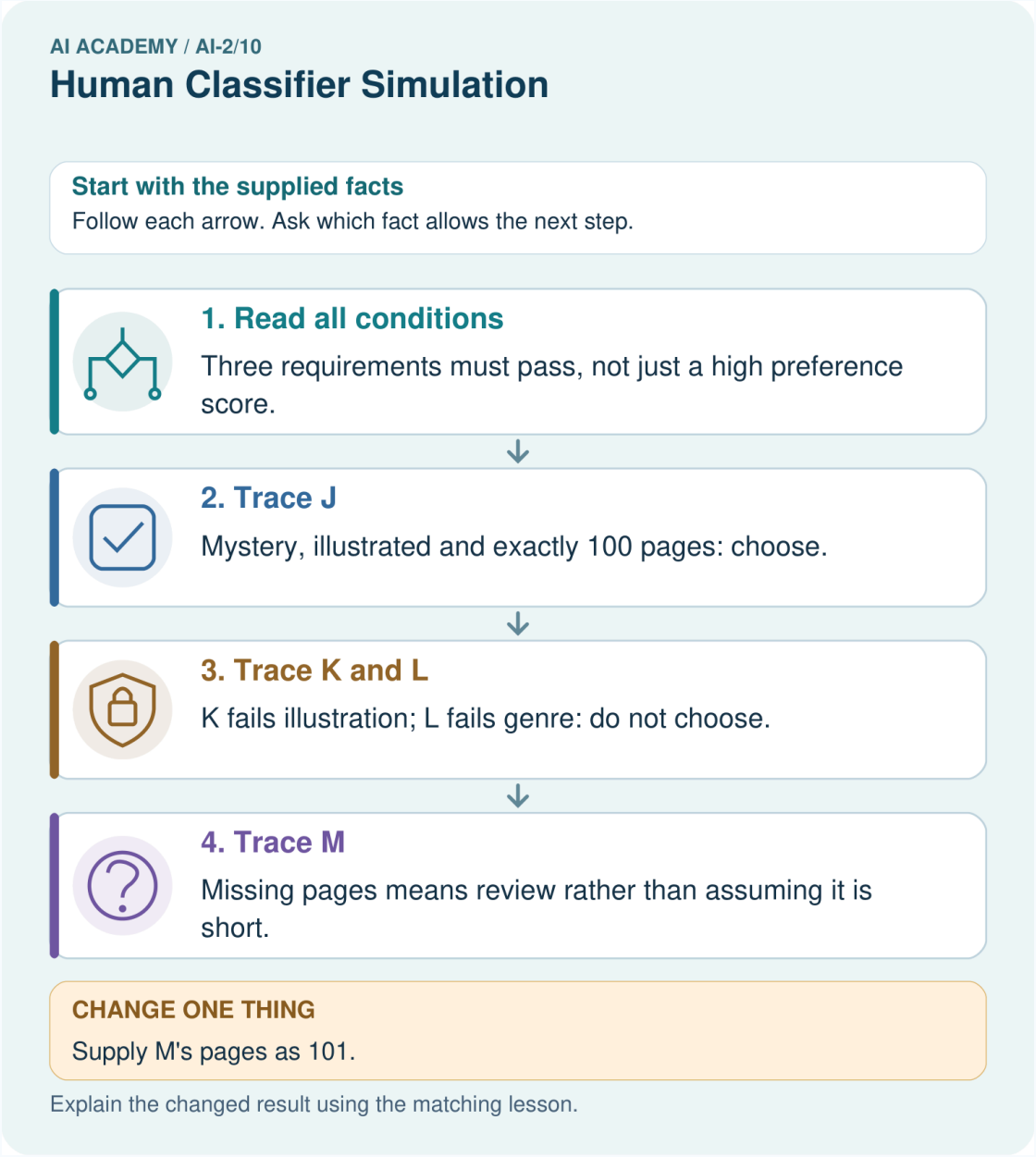
A closer explanation

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Human Classifier Simulation

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Read all conditions	Three requirements must pass, not just a high preference score.
2. Trace J	Mystery, illustrated and exactly 100 pages: choose.
3. Trace K and L	K fails illustration; L fails genre: do not choose.
4. Trace M	Missing pages means review rather than assuming it is short.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

No prediction records a limit of the procedure

> It is an outcome to count, not a third tower-size category.

Separate predicted category from human action

The **classifier** reports a **prediction** or **no prediction**. It does not decide what a person must do with a tower.

A separate classroom procedure might ask someone to remeasure a rejected **input** or compare a **prediction** with a recorded label. Write that action separately from the predicted category.

Keep rejected inputs in the system record

No prediction is not a third tower size. It records that this procedure did not assign short or tall.

Count and investigate such records when evaluating the whole system; hiding them could make a system look more useful than it is.

Challenge input-passed therefore label-correct

> **A classifier accepts 9 cm and predicts tall. A learner says the input passed, so the label is definitely right. Correct the claim. Then explain why reporting short for a blank height would violate the Tower1 procedure**

▼ **Earlier wording, alternatives and picture transcripts**

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: A classifier remembers the correct label for every possible future item. Strong move: locate evidence, change one cause, and retest.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Give your procedure to a partner who follows only the text

> Find unclear branches, then retrace affected records.

Write steps with explicit branch labels

Write the procedure as numbered steps or connected boxes with labelled branches. Give it to a partner who follows only what you wrote.

The partner should record each check and stop point, including equality at the threshold and at both accepted range endpoints.

Trace five supplied partner cases

> **Construct Tower1 in your own words. Have a partner trace 1 cm, 6 cm, 10 cm, 11 cm and a blank height. Record each outcome and any instruction you had to clarify. Explain why the accepted 10 cm case still needs separate evidence of correctness**

Preserve the rule and version during instruction repair

A partner finding an unclear branch gives useful design feedback. Revise the instruction and retrace the affected examples. Keep the model name and threshold visible so a change to the supplied rule is not confused with a correction to an instruction.

Build both outcome paths in W2

Use workbook W2 for the connected example “Constructing a **classifier**”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Repair the numeric-input misconception

- Use the worked case for Constructing a **classifier**. Proposed explanation: “Any numeric **input** must receive a category”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the separate book-case evidence

Why are J and M initially treated differently even though both are illustrated mysteries?

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ Earlier wording, alternatives and picture transcripts

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.

TRACE THE LOGIC

Predict the result. Trace every condition or calculation. Compare expected with actual. Explain the first difference.

SET **evidence**_record PREDICT expected_result RUN one normal case and one **boundary** case COMPARE actual_result with expected_result IF different: locate first cause; change one factor; RETEST EXPLAIN result, limitation, and next safe step

Trace one **input**. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

Practise with less help: Construct both outcome paths

Check your reasoning

Explain the picture

My explanation using a pictured step and the given facts

MAIN LESSON · SECTION 9

Test your written classifier on eight supplied records

> Keep the first trace and the repaired procedure separately.

The rule, contract and check priority

Use Tower1 for this paper investigation. Accepted input: a present whole-number height from 1 through 10, unit cm shown. For an accepted record predict tall at >=6, short otherwise. For a rejected record keep no prediction and the first failed check's reason, then stop that record. Check presence, unit, whole number, then range, in that order. This is a supplied-rule simulation; it does not train a model.

Eight records for the same procedure

Record	Height as supplied
A	1 cm
B	5 cm
C	6 cm
D	10 cm
E	11 cm
F	blank height; unit cm
G	60 mm
H	4.5 cm

Predict, exchange, trace and repair

1. First write predicted procedure outcomes for A–H. Give your W2 instructions to a partner; ask the partner to read each check aloud and record original input, Tower1, acceptance/rejection, outcome and rejection reason. Label these observations partner paper traces. Do not supply hidden corrections while the partner follows the first version. If a record reaches two outcomes or a blank becomes zero, preserve that trace, repair the instruction and retrace all eight records. Compare execution with the expected procedure outcomes; no external truth labels are supplied, so do not report a model accuracy score.

Keep the evidence table and learning note

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** boundary
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** missing
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** fresh transfer
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

What did the hardest test teach you that the normal test did not?

> **Write or draw your reasoning and evidence.**

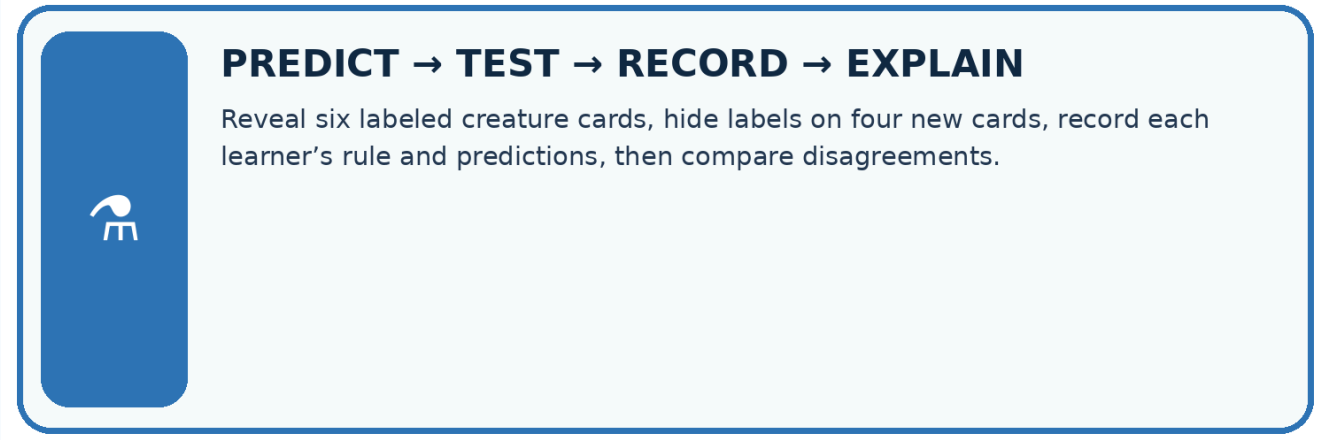
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



The main investigation uses the eight supplied Tower1 records and a partner-operated written procedure. The earlier creature-card idea is an optional alternative only if a teacher supplies a complete labelled set and declared comparison policy; no creature-model observations are supplied here.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: the science-book rule has a strict boundary

> Trace the selected new case before consulting the key.

The independent rule and supplied records

A **written rule** selects science books with fewer than 140 pages. Trace science/139, science/140, history/90 and missing-genre/90. Policy: missing required information means request it.

Your first attempt and help record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback after the attempt

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Changed independent case

Week 10 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A human **classifier** uses sports topic means sports shelf. Classify a sports book and an unreadable cover. State what **evidence** is missing in the second case.

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a **boundary** or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Turn score thresholds into visible boundaries such as recommend / maybe / not now.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Transfer practice from the original programme

Recommend a book only if genre is science AND pages<=150. Classify science/150, science/151, mystery/90 and missing-genre/80. State a missing-data policy.

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

MAIN LESSON · SECTION 11

Add a transparent eligibility stage to the recommender

> Expose failed and unresolved requirements beside the decision.

This week's filtering contribution

This week's portfolio contribution is a transparent eligibility filter. Write the required genre, illustration and page conditions, the missing-field outcome, and a reason for each rejection. A score threshold or trained preference model is a later design choice; do not invent scores to complete this week. Eligibility alone does not establish reader enjoyment.

Build a transparent filtering stage for the recommender, storing each failed or unresolved requirement in its explanation.

Keep system evidence and human choice visible

SYSTEM MAP Reader need → allowed **input**s → transparent rule or score → ranked candidates → explanation → test record → **human decision**

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under **human control**?

Write or draw your reasoning and evidence.

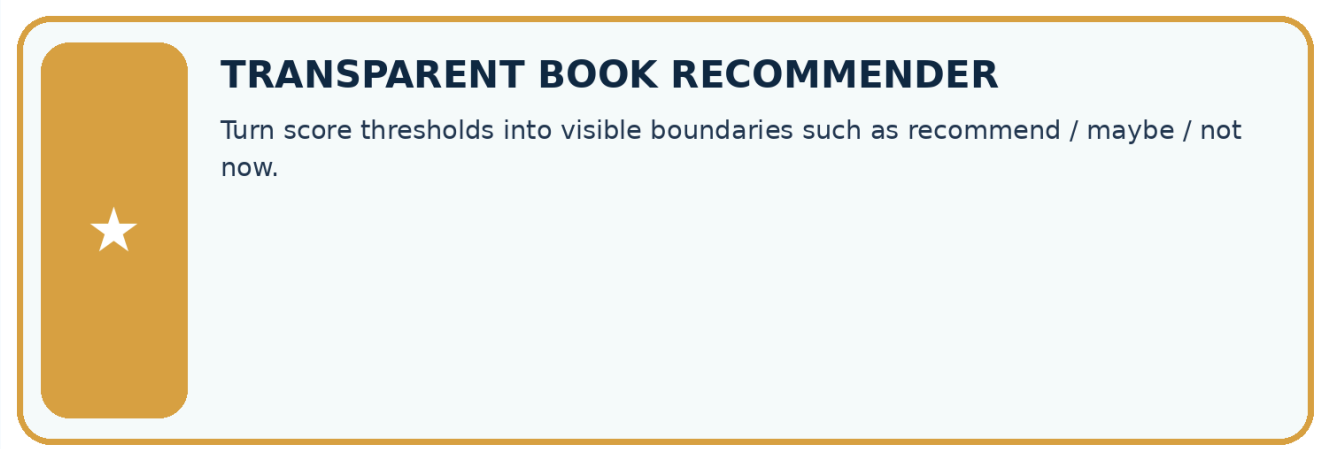
RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse a ranking with a fact about a person.

▼ **Earlier wording, alternatives and picture transcripts**

CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



THIS WEEK'S CHECKPOINT Turn score thresholds into visible boundaries such as recommend / maybe / not now.

Evidence for your project

MAIN LESSON · SECTION 16

Finish: explain acceptance, prediction and correctness separately

> State what the procedure checked and what evidence it still lacks.

Your explanation without notes

- Close your notes. Explain this goal in your own words: Construct a **classifier** with a stated **input** format, selected rule and recorded **output**

My deciding step, evidence and one remaining question

Choose a changed case for return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Your quick word guide**

REFERENCE · SECTION 5

Your quick word guide

> Use the words to explain a particular step, record or limit.

Core vocabulary

Word	My own example	Not this / boundary
classifier	A rule or model that assigns an input to a category.	
boundary	A dividing condition between categories, actions or allowed behaviours.	
training example	An input and any associated label used to fit a model.	
prediction	An estimated category, value or outcome that must be distinguished from established fact.	

Terms for the connected explanation

Word	Meaning
Input contract	The format and conditions an input must satisfy for a procedure to accept it
Validation	Checking whether an input satisfies the stated contract

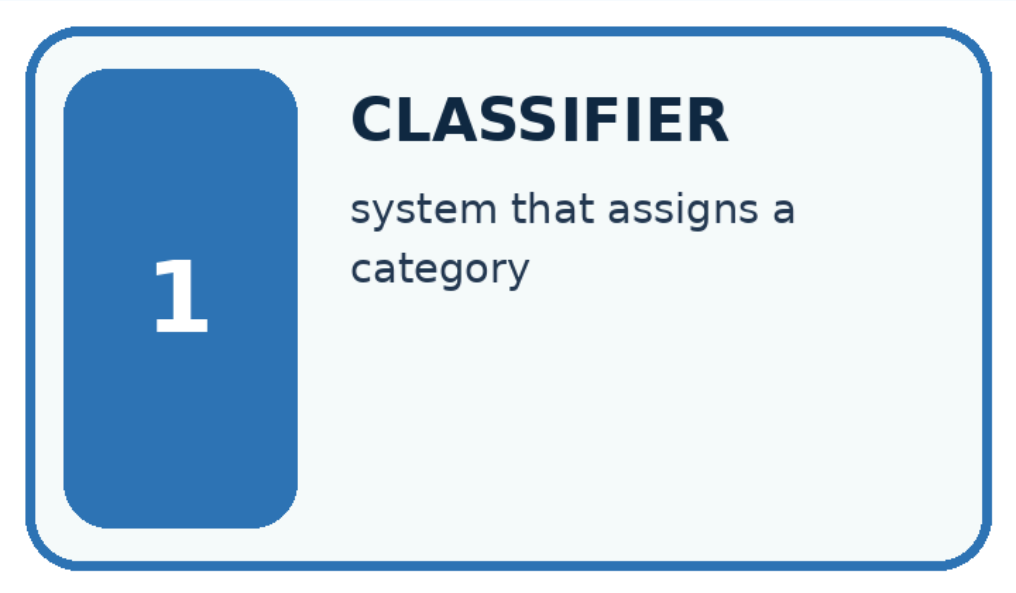
Word	Meaning
No prediction	An explicit outcome when the classifier does not assign a category
Model version	A name that identifies the particular model used for a prediction

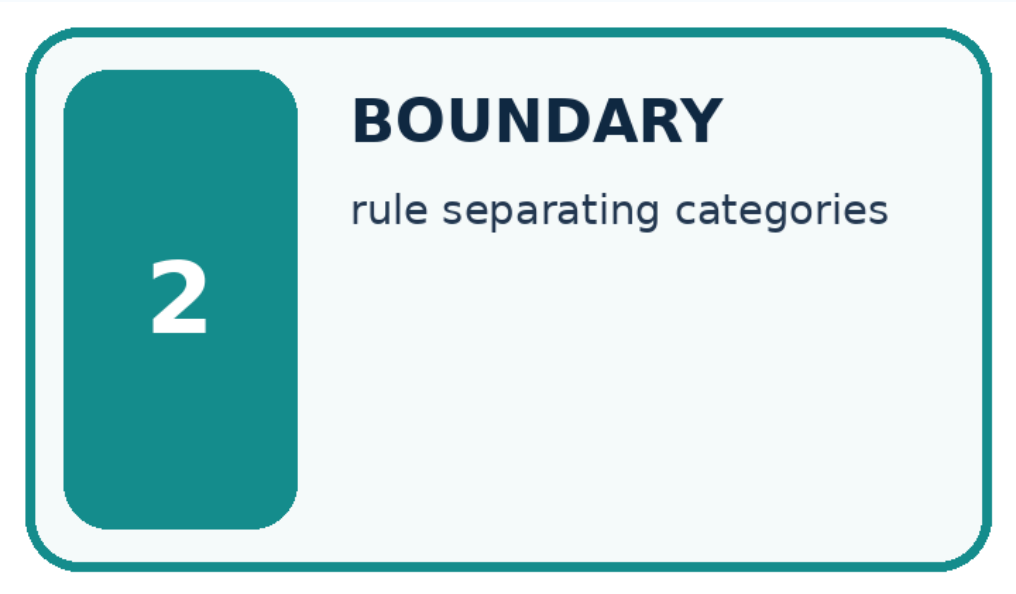
▼ Earlier wording, alternatives and picture transcripts

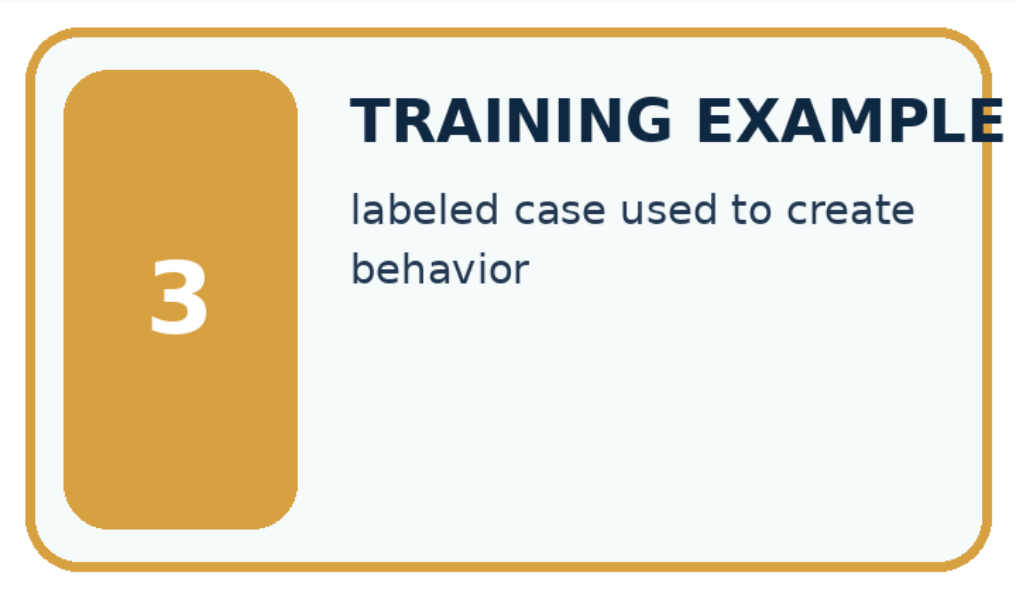
VOCABULARY LAB

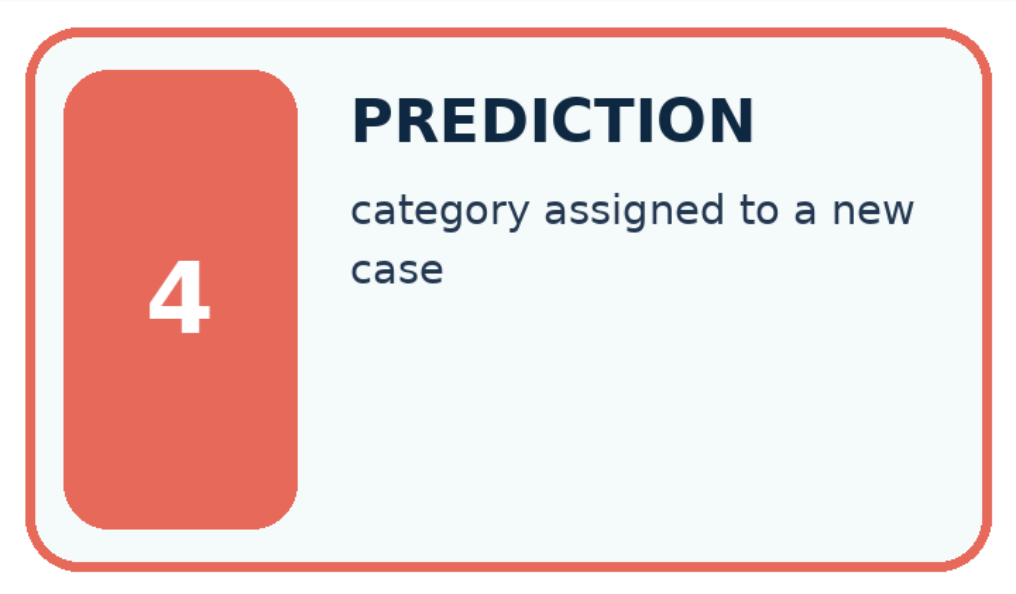
Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.




2
BOUNDARY
rule separating categories


3
TRAINING EXAMPLE
labeled case used to create behavior



Words used in the closer explanation

▼ **Choose a stronger thinking challenge**

REFERENCE · SECTION 10

Choose a stronger thinking challenge

> Choose the next useful challenge rather than treating the whole ladder as compulsory.

Pick your next step

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.
- 3 • APPLY: Solve a different example independently.
- 4 • CHALLENGE: Find a **boundary** case, counterexample, or unfair outcome.

- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

- My strongest **evidence** of independent understanding:

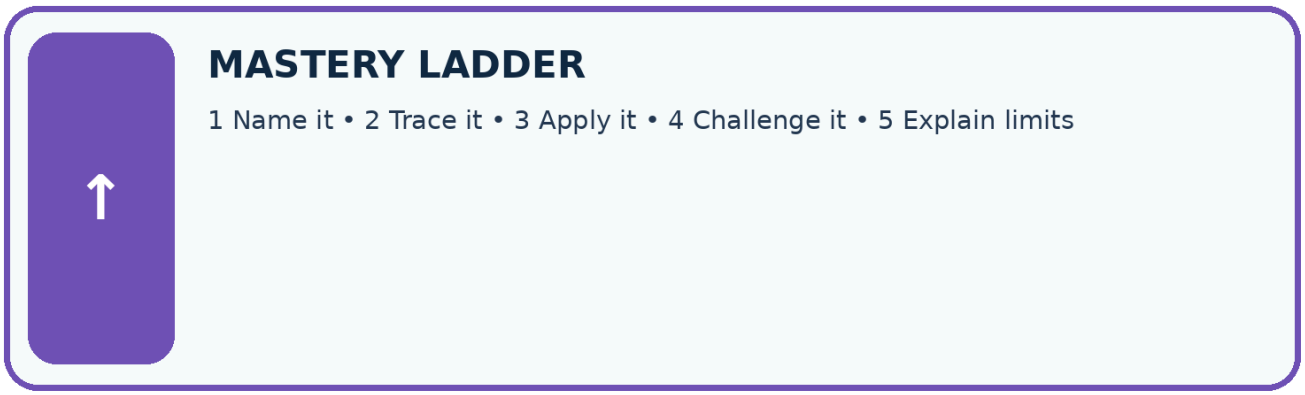
Write or draw your reasoning and evidence.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



▼ Reference: check your explanation

Reference: check your explanation

> Use after practice. Changed independent case is the selected independent case.

Check the evidence and its limit

- 1. My fresh example, trace, calculation, or design:
- 2. My **evidence** that it works (include a **boundary** or missing-data case):
- 3. One limitation, fairness concern, or situation needing human review:

> **Write or draw your reasoning and evidence.**

> **Write or draw your reasoning and evidence.**

Self-check and record

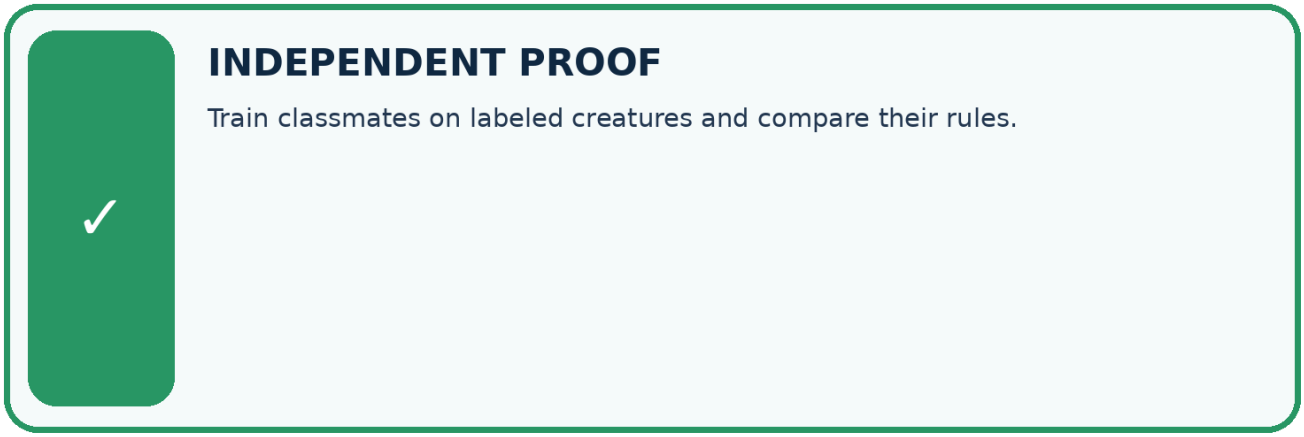
SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

▼ **Earlier wording, alternatives and picture transcripts**

INDEPENDENT MASTERY PROOF

Checklist for explaining this week's idea

Use after practice. Changed independent case is the selected independent case.



EXPLANATION CHECK Make input checks and stops explicit, and distinguish a computed prediction from evidence that it is correct.

Write or draw your reasoning and evidence.

▼ **Extra practice: Both conditions must pass**

REFERENCE · SECTION 13

Extra practice: Both conditions must pass

> Try the question before opening its worked explanation.

Read the supplied case and question

Recommend if adventure AND pages<150; otherwise Other.

Trace A=(adventure,120), B=(adventure,180), C=(science,90).

Your explanation

- Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

▼ Check after trying

A is Recommend. B fails length; C fails genre. Show both conditions.

Evidence or stage	Value or result
Adventure, 120 pages	Genre yes, length yes → Recommend
Adventure, 180 pages	Genre yes, length no → Other
Science, 90 pages	Genre no, length yes → Other

▼ Earlier wording, alternatives and picture transcripts

Week 10 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Both conditions must pass

Recommend if adventure AND pages<150; otherwise Other.

Trace A=(adventure,120), B=(adventure,180), C=(science,90).

A is Recommend. B fails length; C fails genre. Show both conditions.

▼ **Reference: original lesson checklist**

REFERENCE · SECTION 15

Reference: original lesson checklist

> Use an additional checklist when it helps inspect your explanation.

Targets and evidence

By the end, I can	How I will show it
Construct a classifier with a stated input format, selected rule and recorded output	Explain the deciding step, show a trace or test, and state one limit.
Trace valid, boundary and invalid inputs through different paths	Explain the deciding step, show a trace or test, and state one limit.
Distinguish an input check from evidence that a prediction is correct	Explain the deciding step, show a trace or test, and state one limit.

Keep your route and first prediction

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ **Earlier wording, alternatives and picture transcripts**

Start with this question: A fictional librarian wants illustrated mystery books with at most 100 pages.

▼ **Extra practice: pictures, repair and feedback**

REFERENCE · SECTION 17

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional librarian wants illustrated mystery books with at most 100 pages.
- Which labelled step matches this detail? Missing pages means review rather than assuming it is short.
- 1. Read all conditions
- 2. Trace M
- 3. Trace J

- 4. Trace K and L

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Trace M Missing pages means review rather than assuming it is short. This human simulation lets students inspect every decision without needing a trained model.

Equality matters at 100. Known failures give reasons for rejection; M lacks enough information for the page check. The order should not turn blank into zero or quietly ignore a required field.

Changed-case question

- Changed-case practice: Supply M's pages as 101. Which conclusion is supported by the supplied facts?

- 1. M is now rejected rather than awaiting page review.

- 2. We can decide the result without examining the changed **input** or the procedure.

- 3. This one example guarantees correct results for every future **input**.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed case**

After your attempt, compare with the supported explanation: M is now rejected rather than awaiting page review. The required field is known but fails the maximum-length condition.

Repair and save your work

Review Features and Distractors (ai-2/9). Goal: Choose recorded properties that preserve distinctions needed for a stated task. Short practice: Does this one pair prove the **classifier** always ignores subject?

Explain a narrower claim.. Return to this week and explain what you changed.

Review Rows Columns and Data Types (ai-2/7). Goal: Specify a table schema. Short practice: Which two rows originally fail, and do they fail for the same reason?. Return to this week and explain what you changed.

Review Trace Tables (ai-2/4). Goal: Trace variable values before and after each loop iteration. Short practice: Move print inside the original loop. What sequence is displayed?.

Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

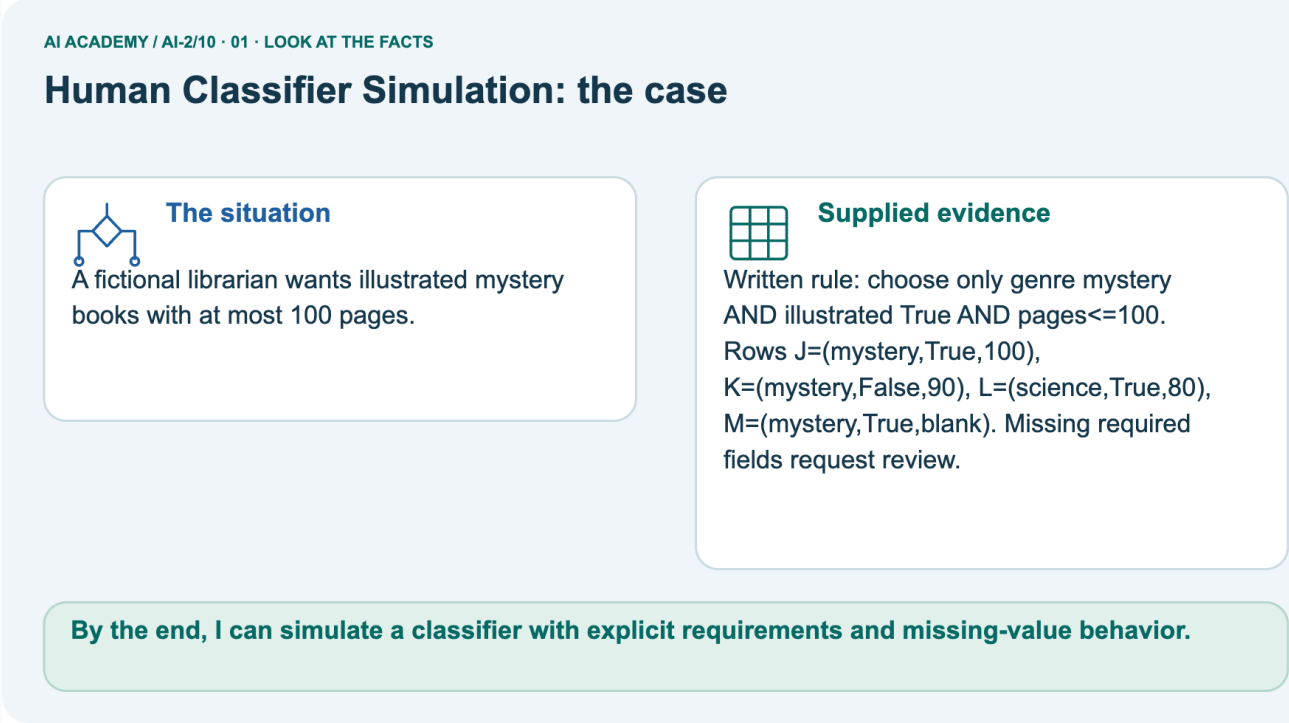
▼ **Reference: the six-picture journey**

REFERENCE · SECTION 18

Reference: the six-picture journey

> Revisit the picture that serves the current question; key diagrams already sit beside the main explanation.

01. Human Classifier Simulation: the case



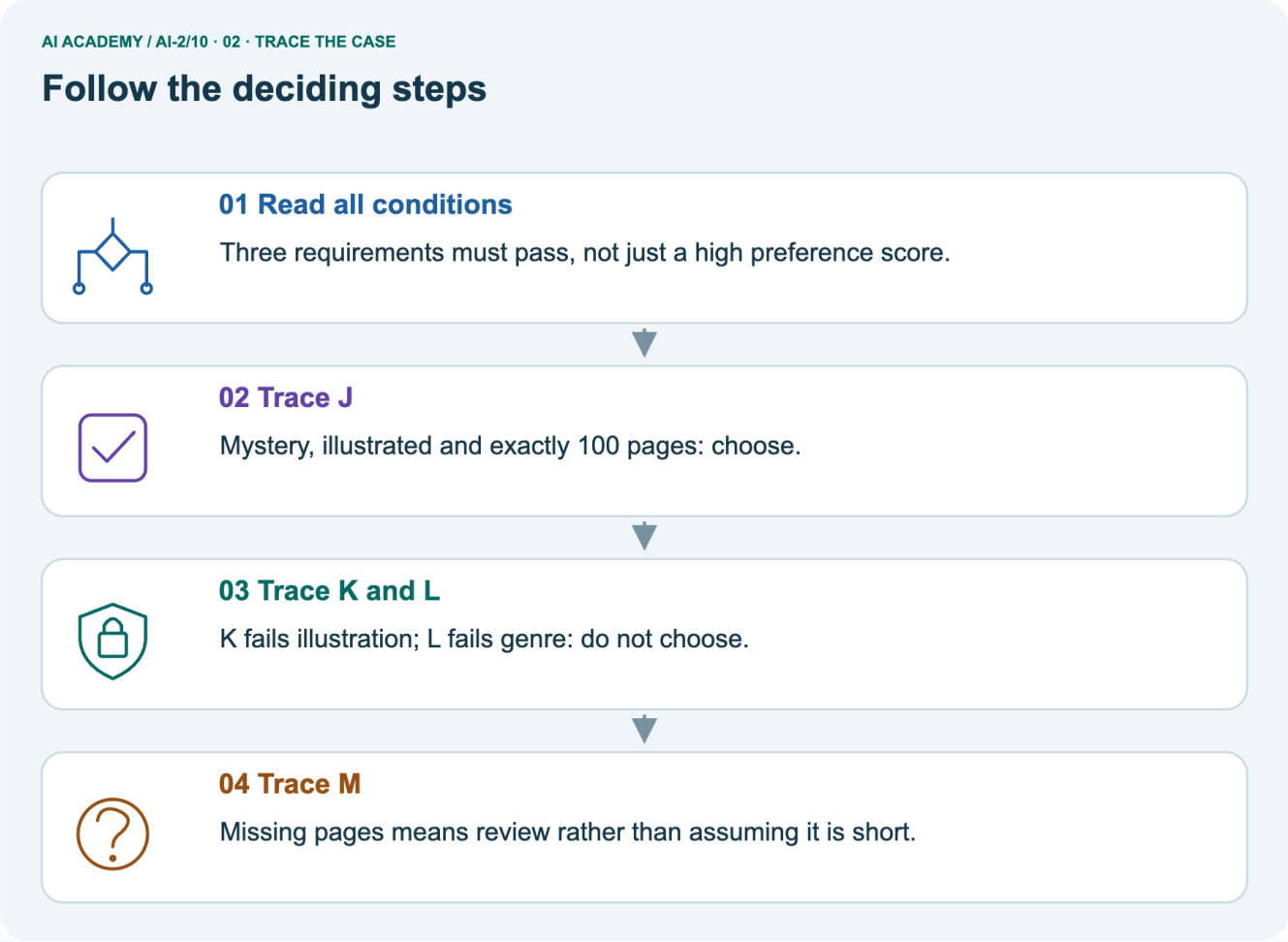
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional librarian wants illustrated mystery books with at most 100 pages. Supplied facts: **Written rule**: choose only genre mystery AND illustrated True AND pages<=100.

Rows J=(mystery,True,100), K=(mystery,False,90), L=(science,True,80), M=(mystery,True,blank). Missing required fields request review.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

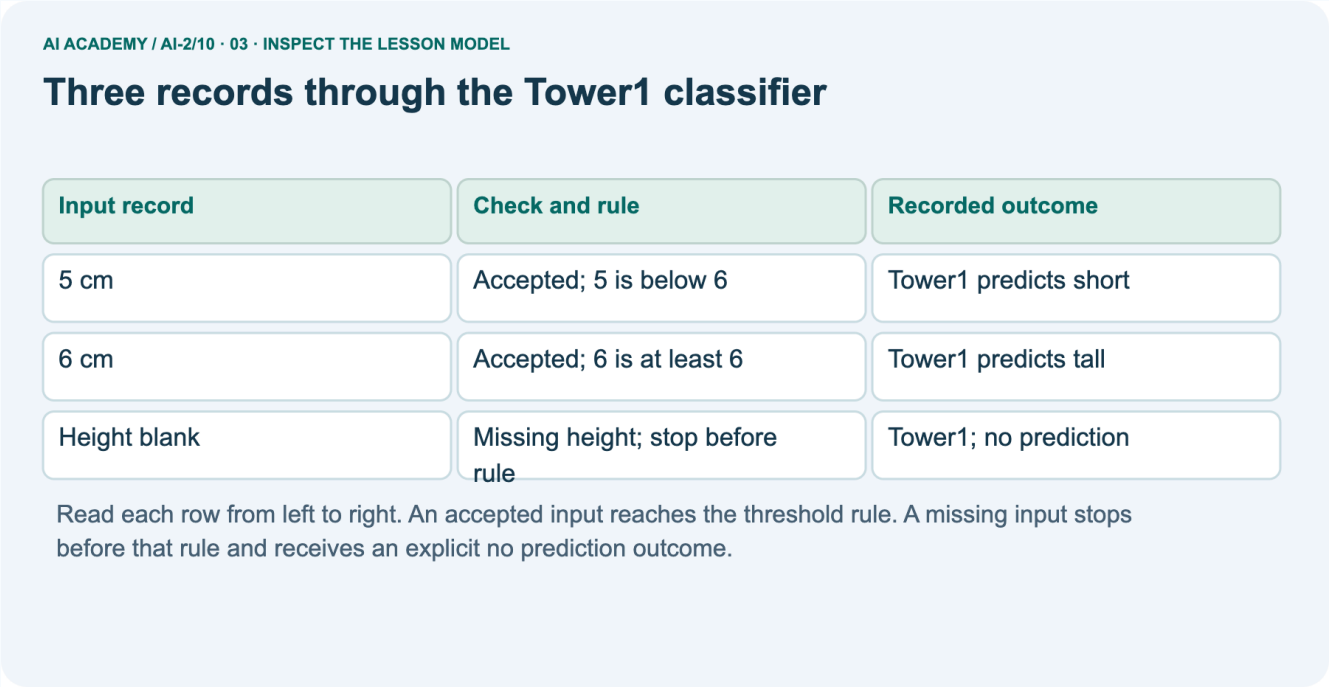


Follow the labelled picture with the matching explanation. Open the vector diagram

Read all conditions: Three requirements must pass, not just a high preference score. → Trace J: Mystery, illustrated and exactly 100 pages: choose. → Trace K and L: K fails illustration; L fails genre: do not choose. → Trace M: Missing pages means review rather than assuming it is short.

Point to each numbered stage and explain which recorded **input** it uses.

03. Three records through the Tower1 classifier



AI ACADEMY / AI-2/10 · 03 · INSPECT THE LESSON MODEL

Three records through the Tower1 classifier

Input record	Check and rule	Recorded outcome
5 cm	Accepted; 5 is below 6	Tower1 predicts short
6 cm	Accepted; 6 is at least 6	Tower1 predicts tall
Height blank	Missing height; stop before rule	Tower1; no prediction

Read each row from left to right. An accepted input reaches the threshold rule. A missing input stops before that rule and receives an explicit no prediction outcome.

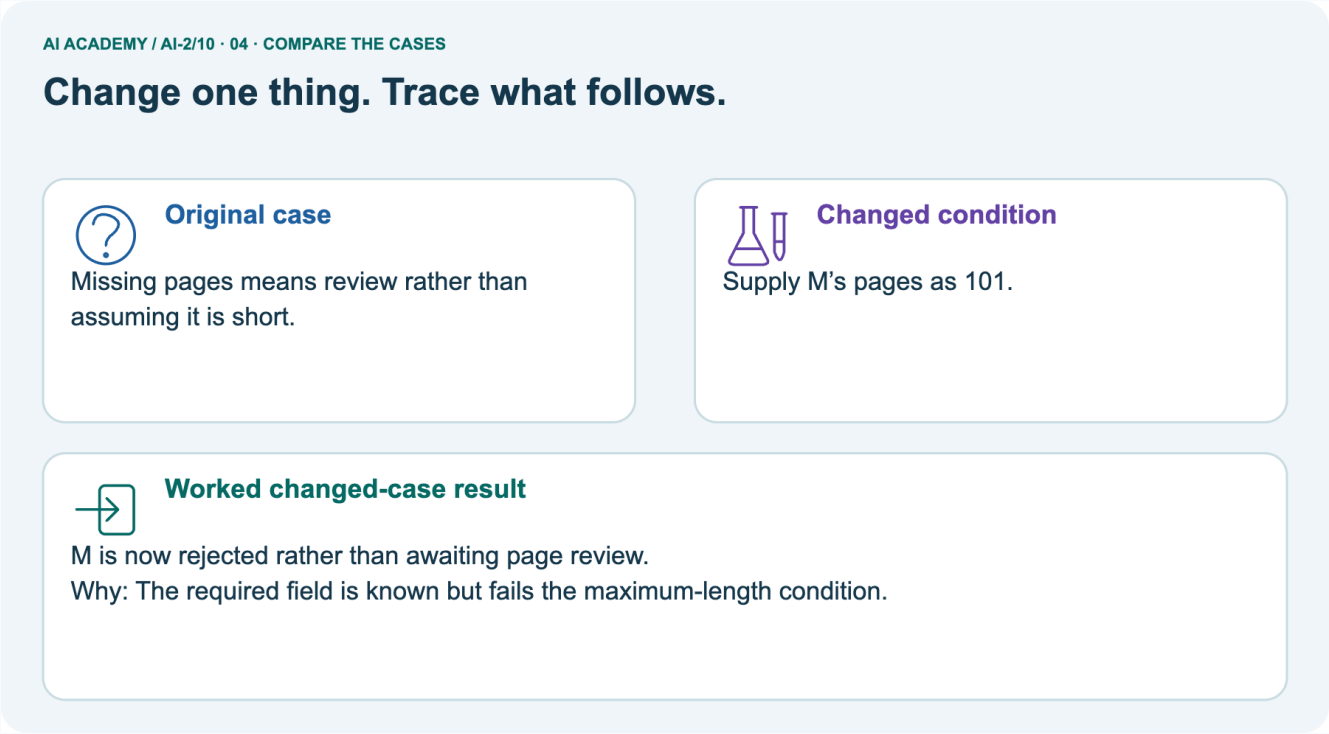
Follow the labelled picture with the matching explanation. Open the vector diagram

Three records through the Tower1 **classifier**. 5 cm | Accepted; 5 is below 6 | Tower1 predicts short. 6 cm | Accepted; 6 is at least 6 | Tower1 predicts tall.

Height blank | Missing height; stop before rule | Tower1; **no prediction**

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



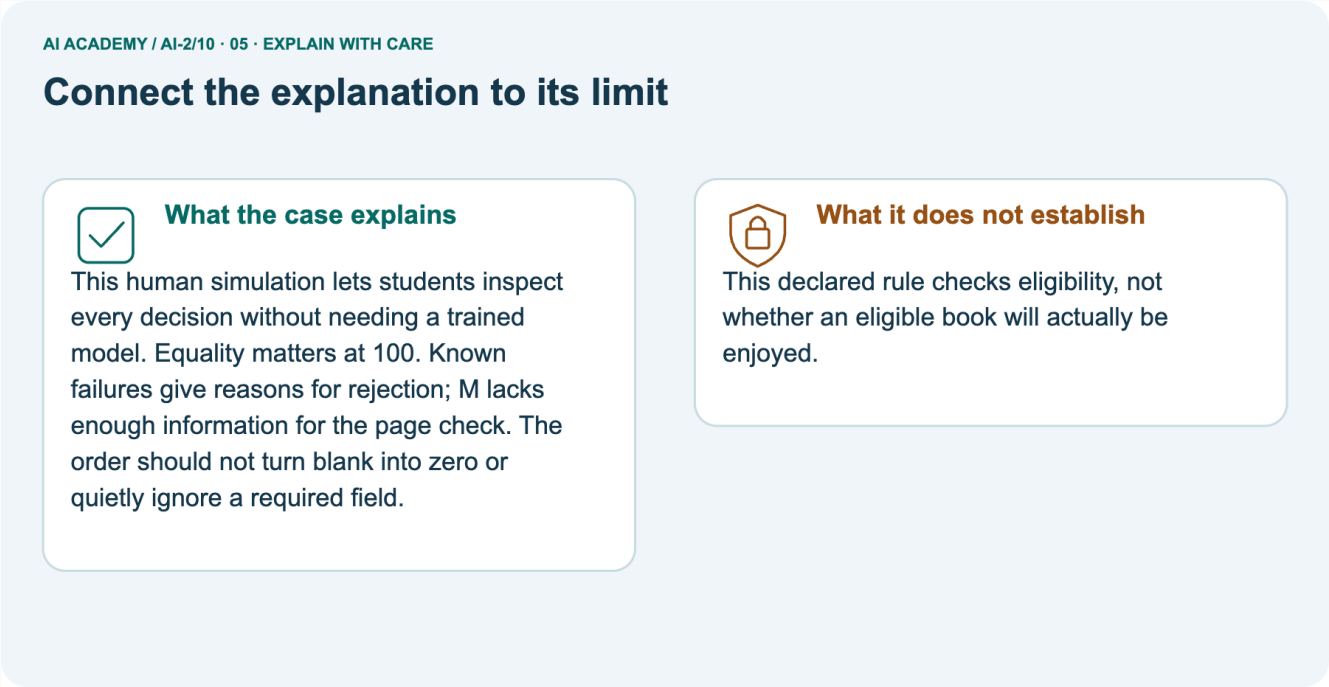
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Missing pages means review rather than assuming it is short. Changed condition: Supply M's pages as 101. Expected result: M is now rejected rather than awaiting page review.

Reason: The required field is known but fails the maximum-length condition.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



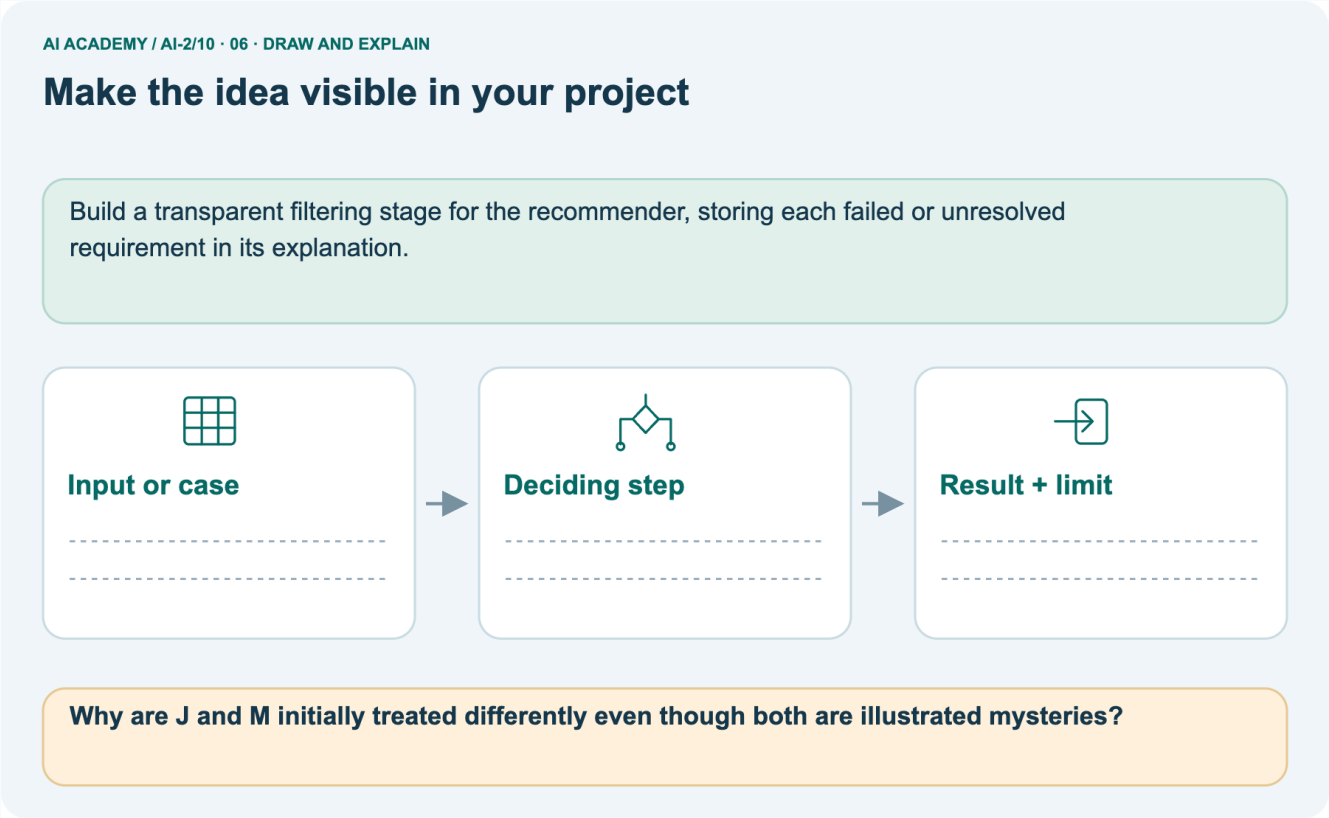
Follow the labelled picture with the matching explanation. Open the vector diagram

This human simulation lets students inspect every decision without needing a trained model. Equality matters at 100. Known failures give reasons for rejection; M lacks enough information for the page check.

The order should not turn blank into zero or quietly ignore a required field. Limit: This declared rule checks eligibility, not whether an eligible book will actually be enjoyed.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Build a transparent filtering stage for the recommender, storing each failed or unresolved requirement in its explanation. Draw your **input**, deciding step, result and limit.

Supported question: Why are J and M initially treated differently even though both are illustrated mysteries?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Human Classifier Simulation: the case

02. Follow the deciding steps

03. Three records through the Tower1 classifier

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.